



SHEETS / PLATES

SUMMARY OF THE MAIN ASTM STANDARDS GENERALLY USED FOR SHEETS / PLATES

ASTM	Grade	Chemical requirements percent (%)										Mechanical requirements				
		C	Mn	P	S	Si	Ni	Cr.	Mo	Cu	Others	Tensile Strength mini-MPa	Yield Strength mini-MPa	Elong mini %	Hardness Brinell	Rockwell
A240	304	0.08	2.00	0.045	0.030	0.75	8.00-10.5	18.00-20.0				515	205	40	201	92
	304L	0.03	2.00	0.045	0.030	0.75	8.00-12.0	18.00-20.0				485	170	40	201	92
	310	0.08	2.00	0.045	0.030	1.50	19.0-22.0	24.0-28.0				515	205	40	217	95
	316	0.08	2.00	0.045	0.030	0.75	10.0-14.0	16.0-18.0	2.00-3.00			515	205	40	217	95
	316L	0.03	2.00	0.045	0.030	0.75	10.0-14.0	16.0-18.0	2.00-3.00			485	170	40	217	95
	317L	0.03	2.00	0.045	0.030	0.75	11.0-15.0	18.0-20.0	3.00-4.00			515	205	40	217	95
	321	0.08	2.00	0.045	0.030	0.75	9.00-12.0	17.0-19.0			TT>50<0.70	515	205	40	217	95
A387 Class1 Class2	347	0.08	2.00	0.045	0.030	0.75	9.00-13.0	17.0-19.0			Cb+Ta > 100<1.10	515	205	40	201	92
	2	0.05-0.21	0.55-0.80	0.035	0.040	0.15-0.40	0.50-0.80	0.45-0.60				Class 1	Class 2			
	5	0.15	0.30-0.60	0.04	0.030	0.050	4.00-5.00	0.45-0.65				380	230	22	max201HB	max92HRB
	7	0.15	0.30-0.60	0.030	0.030	1.00	6.00-8.00	0.45-0.65				415	205	18	max202HB	max92HRB
	9	0.15	0.30-0.60	0.030	0.030	1.00	8.00-10.0	0.90-1.10				415	205	18	max217HB	max95HRB
	11	0.04-0.17	0.40-0.65	0.035	0.04	0.50-0.80	1.00-1.50	0.45-0.65				415	205	18	max217HB	max95HRB
	12	0.04-0.17	0.40-0.65	0.035	0.04	0.15-0.40	0.80-1.15	0.45-0.60				415	240	22	max217HB	max95HRB
	21	0.04-0.17	0.30-0.60	0.035	0.035	0.50	2.75-3.25	0.90-1.10				380	230	22	max217HB	max95HRB
	22	0.05-0.17	0.30-0.60	0.035	0.035	0.50	2.00-2.50	0.90-1.10				415	205	18	max201HB	max92HRB
	55	0.22	0.90	0.035	0.04	0.15-0.40						415	205	18	max201HB	max92HRB
	60	0.27	0.90	0.035	0.04	0.15-0.40						380-515	205	27		
A 515	65	0.31	0.90	0.035	0.04	0.15-0.40						415-550	220	25		
	70	0.33	1.20	0.035	0.04	0.15-0.40						450-585	240	23		
	55	0.20	0.60-1.20	0.035	0.04	0.15-0.40						485-620	260	21		
	60	0.23	0.85-1.20	0.035	0.04	0.15-0.40						380-515	205	27		
A 516	65	0.26	0.85-1.20	0.035	0.04	0.15-0.40						415-550	202	25		
	70	0.28	0.85-1.20	0.035	0.04	0.15-0.40						450-585	240	23		
	Class 1	0.24	0.70-1.35	0.035	0.040	0.15-0.40	0.25 max	0.80 max	0.35 max			485-620	345	22		
	Class 2	0.24	0.70-1.35	0.035	0.040	0.15-0.40	0.25 max	0.80 max	0.35 max			550-690	415	22		

IS-2002-62 STEEL PLATES FOR BOILERS

Designation	Chemical Composition					Tensile Test		Elongation	
	c	Si	P	S	max	Tensile strength Mpa	Yield Strength Mpa	Test Piece	%min
IS 2002-1	0.18	0.10-0.35	0.040	0.040	0.040	362-442	540	5.65/Sc 4/Sc	26 30
IS 2002-2A	0.20	0.10-0.35	0.050	0.050	0.050	412-491	491	5.60/Sc 4/Sc	25 29
IS 2002-2B	0.22	0.10-0.35	0.050	0.050	0.050	510-608	491	5.65/Sc 4/Sc	20 24

IS-2062-92 STEEL FOR GENERAL STRUCTURAL PURPOSES

Grade Designation	% Chemical Composition					Tensile strength (Min)	Yield Strength (Min)		%El.in gauge length 5.56/Sc	Bend Test	Std test Piece Charpy V Notch Impact Energy Joule min
	C	MN	S	P	Si		<20 min	>40 min			
A FE410 WA	0.23	1.5	0.050	0.050	-	0.42	41.8	250	230	3t	-
B FE410 WB	0.22	1.5	0.045	0.045	0.40	0.41	41.8	250	230	<25mm 3t for >25mm	2t for 27 3t for >25mm
C FE410 WC	0.20	1.5	0.040	0.040	0.40	0.36	41.8	250	230	2t	27

Formula - Weight of Stainless Steel Sheets/Plates = Length (mm) x Width (mm) x Thickness (mm) x 7.86 = Kg/Sheet.